

Technical Document #98: Data Engineering - Comprehensive Overview

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Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Data quality ensures data is accurate, complete, and consistent.
- Data governance establishes policies and procedures for managing data assets.
- Data lakes store raw data in its native format.